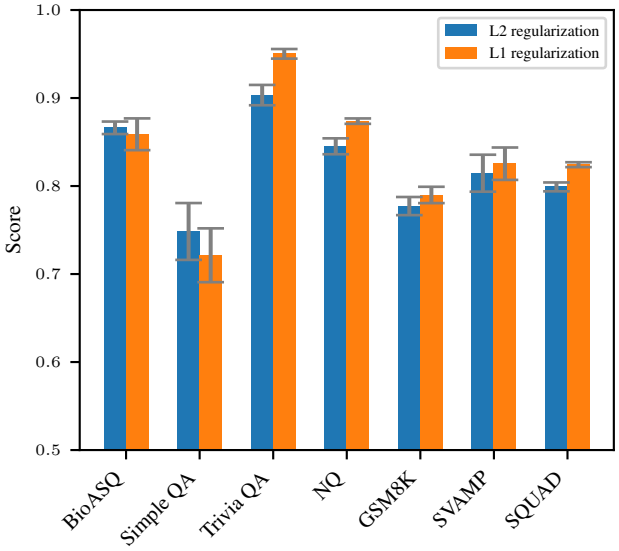

Figure 18. AUROC of linear probes at the stop token of the output, trained with either L1 or L2 regularization.

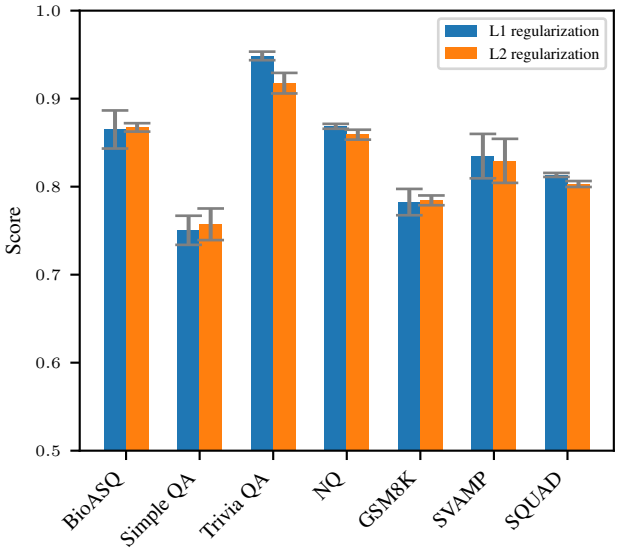

Figure 19. AUROC of linear probes at the token before the stop token of the output, trained with either L1 or L2 regularization.

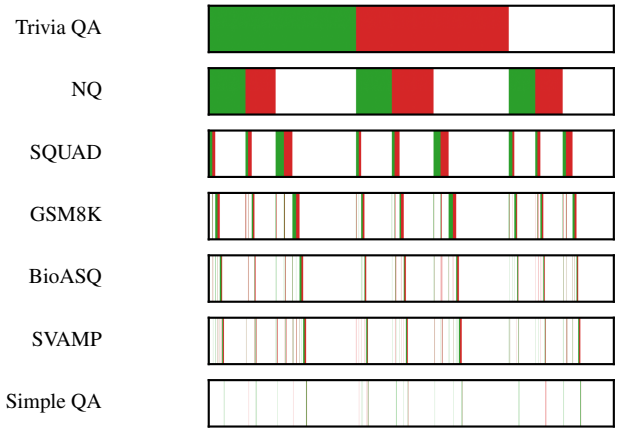

Figure 20. Signed support of sparse probes trained on different datasets at the stop token of the output, using L1 regularization at layer 28. Each row represents one probe trained on the corresponding dataset (y-axis labels). The x-axis shows the 3584 dimensions of the hidden state vector. Green indicates positive coefficients, red indicates negative coefficients, and white indicates zero coefficients (sparsity). Dimensions are sorted by sparsity level across all datasets, with the least sparse dimensions on the left and the most sparse on the right.

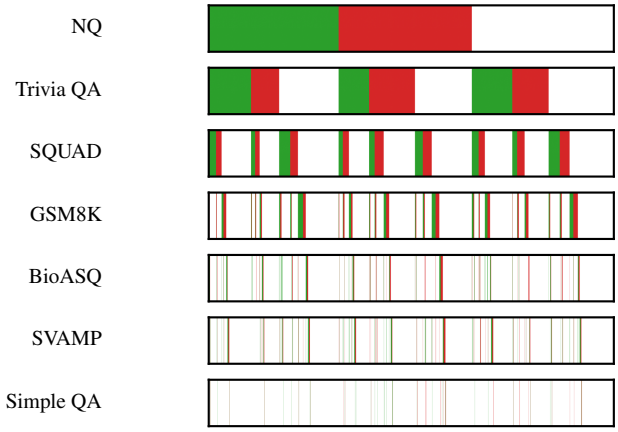

Figure 21. Signed support of sparse probes trained on different datasets at the token before the stop token of the output, using L1 regularization at layer 28. Each row represents one probe trained on the corresponding dataset (y-axis labels). The x-axis shows the 3584 dimensions of the hidden state vector. Green indicates positive coefficients, red indicates negative coefficients, and white indicates zero coefficients (sparsity). Dimensions are sorted by sparsity level across all datasets, with the least sparse dimensions on the left and the most sparse on the right.

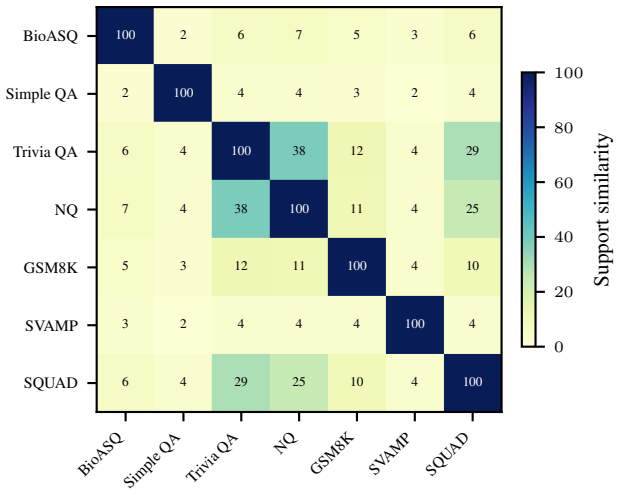

Figure 22. Support overlap between sparse probes trained on different datasets. Darker colors indicate higher overlap percentages. Task pairs with $> 30\%$ overlap (TriviaQA, NQ, SimpleQA) correspond to successful cross-task generalization, while most pairs show $< 15\%$ overlap, explaining generalization failure.

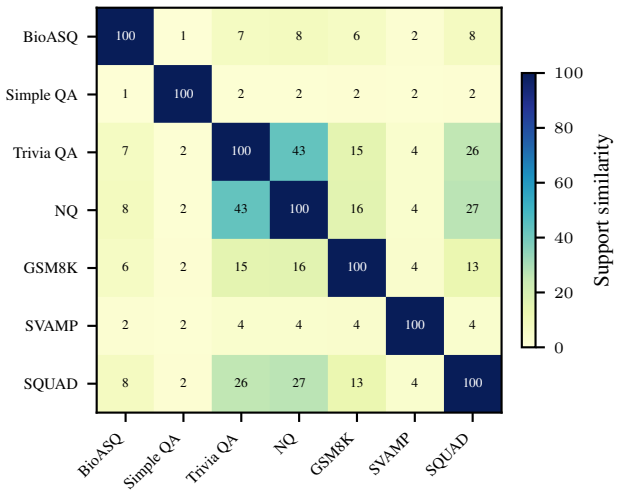

Figure 23. Support overlap between sparse probes trained on different datasets. Darker colors indicate higher overlap percentages. Task pairs with $> 30\%$ overlap (TriviaQA, NQ, SimpleQA) correspond to successful cross-task generalization, while most pairs show $< 15\%$ overlap, explaining generalization failure.

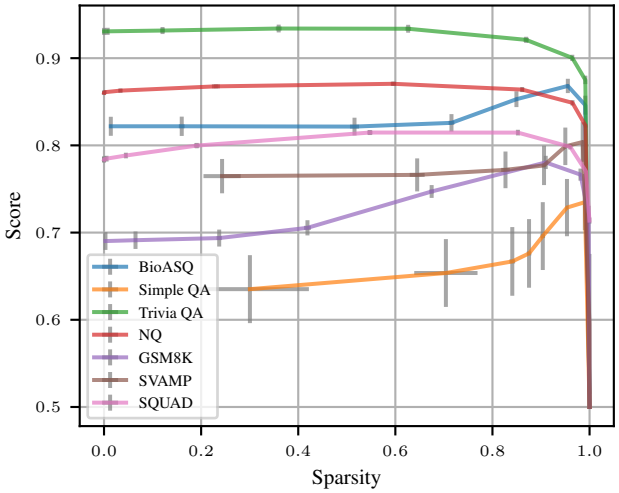

Figure 24. AUROC of probes trained using L1 regularisation as a function of the sparsity level on the stop token of the output.

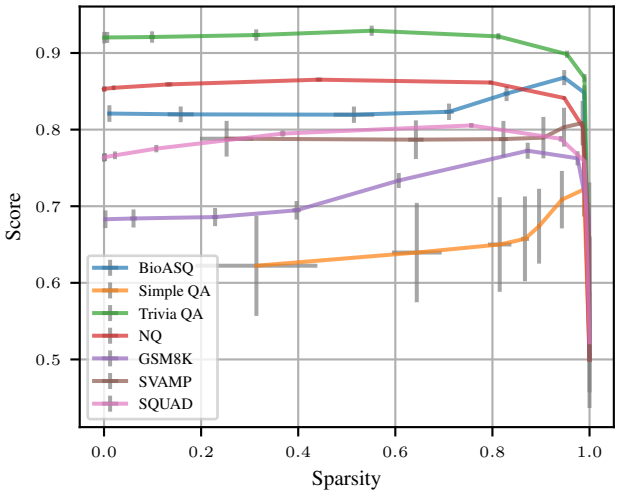

Figure 25. AUROC of probes trained using L1 regularisation as a function of the sparsity level on the token before the stop token of the output.

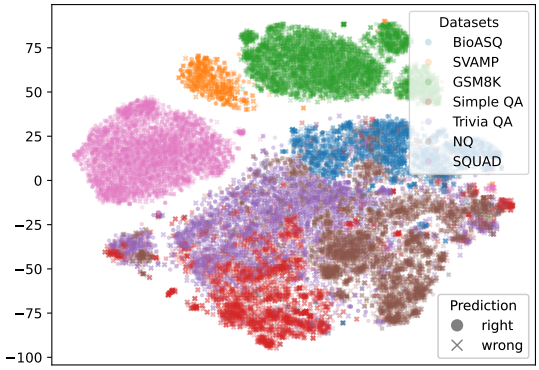

Figure 26. t-SNE plots of the hidden space at layer 28 at the stop token of the output.

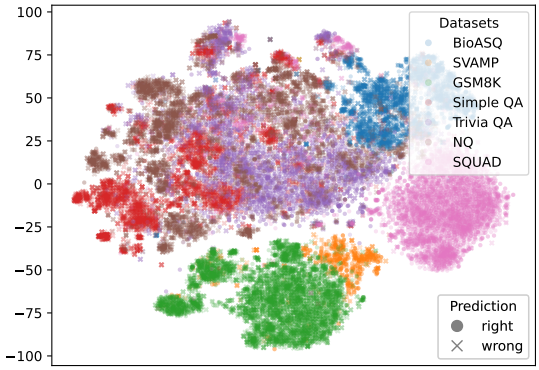

Figure 27. t-SNE plots of the hidden space at layer 28 at the token before stop token of the output.